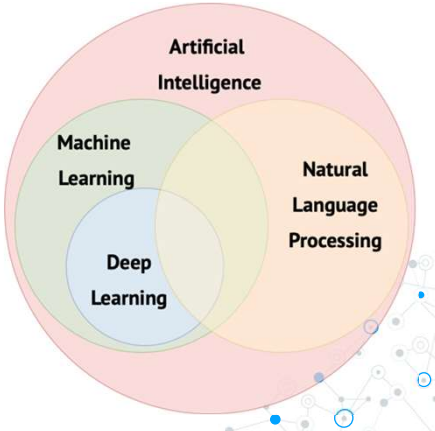
INTRODUCTION TO NATURAL LANGUAGE PROCESSING

Dr. Shailesh S



Where NLP fits in..

Notable **Successes** of Neural NLP Models

Speech Recognition

Machine Translation

Smart Assistants

Information Retrieval

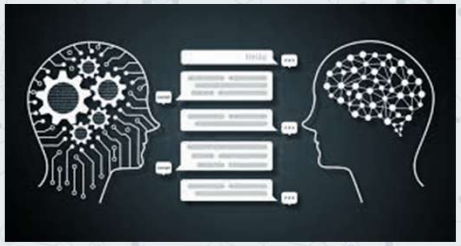
Story Generation

.....

Other NLP applications...

- **prediction of diseases** from electronic health records
- extracting **customers choices and their decision drivers**
- **identifying fake news and hate speech** in social media posts
- **stock market prediction** by tracking news, reports, comments about possible mergers between companies
- In talent recruitment, **identifying the skills of potential hires** and also spotting prospects
- **chatbot therapist** helping people with anxiety and other disorders.
- organizations using chatbots will cut customer service costs by $80 billion in 2025

What is Natural Language Processing ?



- Designing/building computational models to *understand* and *generate* human languages to get some useful task done

What is NLP??

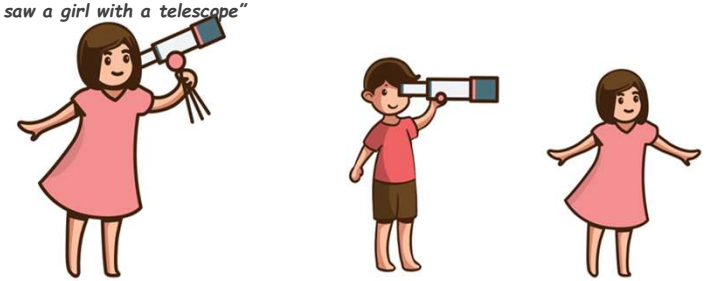
Natural language processing (NLP)- A branch of Artificial Intelligence that gives machines the ability to understand natural human langauge.

billions of text data being generated every day and most of them are unstructured.

Why is NLP challenging ?

- Language is Ambiguous
  - Lexical, Syntactic, Semantic, Pragmatic etc.
- Language has Irregularities like
  - sarcasm,
  - idioms,
  - metaphors
  - and lot more..
- Language keeps evolving
- Linguistic communication is compressed

"I saw a girl with a telescope"



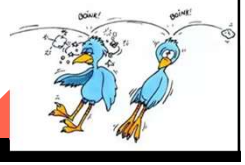
PP-attachment problem : An example of **Syntactic Ambiguity**

Semantic Ambiguity

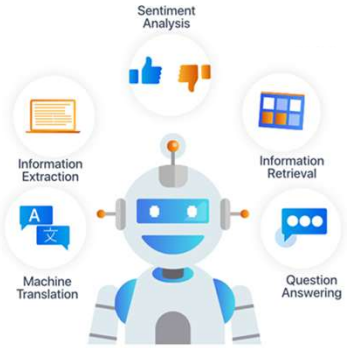
Four different interpretation for a simple sentence !!

Some more example.....

- He has a heart of gold
- Meet me at the bank.
- Kill two birds with one stone.
- It's a piece of cake
- It's raining cats and dogs
- She is a walking encyclopedia.
- He broke my heart.
- A great movie for a sunday nap!



Applications of Natural Language Processing in Different Domains



Missing Text Phenomenon

- We all know what we all know, but machine don't!!

plastic cup	is a [cup made of plastic]
plastic factory	is a [factory that produces plastic]
coffee cup	is a [cup to hold coffee]
coffee machine	is a [machine to make coffee]
computer store	is a [store that sells computers]
neighborhood store	is a [store in the neighborhood]
etc.	etc.

The Missing Text Phenomenon, Again: the case of Compound Nominals

Evolution of NLP over the years

Rule based NLP — 1950 -1990

- Uses a well-defined set of rules
- Not scalable
- Need domain expertise

Statistical NLP — 1990 - 2010

- Uses statistics to learn rules
- Relies on hand-crafted features
- Need domain expertise

Neural NLP — 2010 - Today

- Uses artificial neural networks
- Extracts features from raw data
- No need of domain expertise

Rule - Based NLP System

Think of some useful rules to filter spam emails ??

Rule - Based NLP System

Think of some useful rules to filter spam emails ??

R1: misspelling in company names
R2: overuse of words like “lottery”,”dollars”
R3: Unnecessary urgency
R4: Inappropriate languages
R5:...

Statistical NLP System

Labeled Training data

D1: “send us your password”	**Spam**
D2: “send us your review”	**Spam**
D3: “review your password”	**Ham**
D4: “review us”	**Ham**
D5: “send us password”	**Spam**
D6: “send us your account ”	**Spam**

Convert to probabilities

P(Spam)=4/6	P(Ham)=2/6

Spam	Ham	
2/4	1/2	password
1/4	2/2	review
3/4	1/2	send
3/4	1/2	us
3/4	1/2	your
1/4	0/2	account

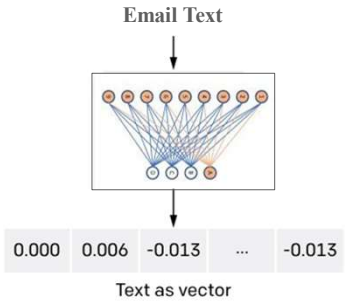
Apply Bayes Theorem

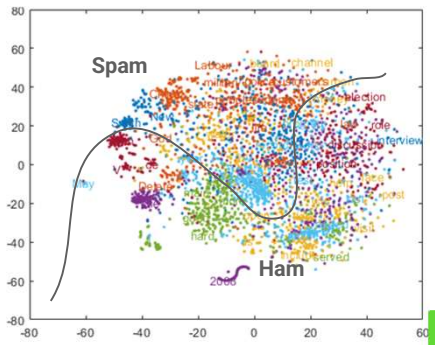
$$P(c \mid X) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c)$$

P(Spam|”review us now”)

Compute the probability of a given review being spam

Neural NLP System





Python Libraries for NLP

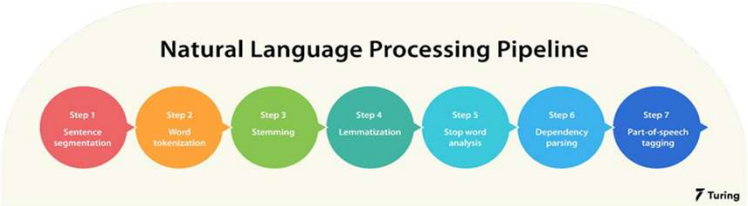
TextBlob

spaCy

GENSIM

NLTK

Common Terminologies



Tokenization

Stemming and Lemmatization

Stemming vs Lemmatization

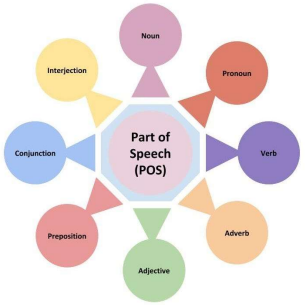
Stop Words

Stop Words

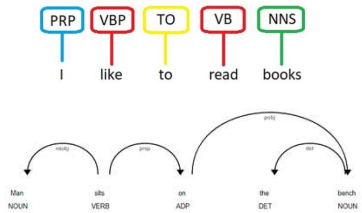
- a
- I
- the
- in
- of
- for
- at
- to
- on
- with
- from

When was the first computer invented?
How do I install a hard disk drive?
How do I use Adobe Photoshop?
Where can I learn more about computers?
How to download a video from YouTube
What is a special character?
How do I clear my Internet browser history?
How do you split the screen in Windows?
How do I remove the keys on a keyboard?
How do I install a hard disk drive?

Part Of Speech(POS)



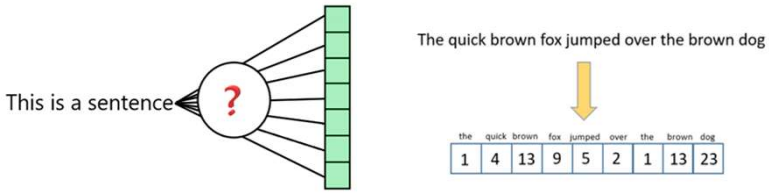
POS Tagging



Named Enity Recognition

contentSkip to site indexPoliticsSubscribeLog InSubscribeLog InToday's PaperAdvertisementSupported ORG byF.B.I. Agent Peter Strzok PERSON , Who Criticized Trump PERSON in Texts, Is FiredImagePeter Strzok, a top F.B.I. GPE counterintelligence agent who was taken off the special counsel investigation after his disparaging texts about President Trump PERSON were uncovered, was fired. CreditT.J. Kirkpatrick PERSON for The New York TimesBy Adam Goldman ORG and Michael S. SchmidtAug PERSON . 13 CARDINAL , 2018WASHINGTON CARDINAL — Peter Strzok PERSON , the F.B.I. GPE senior counterintelligence agent who disparaged President Trump PERSON in inflammatory text messages and helped oversee the Hillary Clinton PERSON email and Russia GPE investigations, has been fired for violating bureau policies, Mr. Strzok PERSON 's lawyer said Monday DATE .Mr. Trump and his allies seized on the texts — exchanged during the 2016 DATE campaign with a former F.B.I. GPE lawyer, Lisa Page — in PERSON assailing the Russia GPE investigation as an illegitimate "witch hunt." Mr. Strzok PERSON , who rose over 20 years DATE at the F.B.I. GPE to become one of its most experienced counterintelligence agents, was a key figure in the early months DATE of the inquiry.Along with writing the texts, Mr. Strzok PERSON was accused of sending a highly sensitive search warrant to his personal email account.The F.B.I. GPE had been under immense political pressure by Mr. Trump PERSON to dismiss Mr. Strzok PERSON , who was removed last summer DATE from the staff of the special counsel, Robert S. Mueller III PERSON . The president has repeatedly denounced Mr. Strzok PERSON in posts on

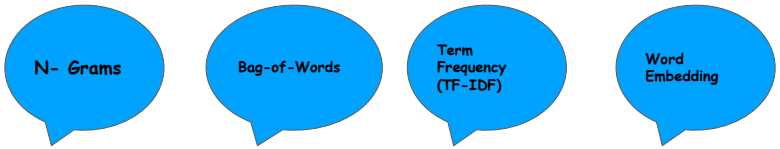
Text Features Extraction



Turning text into vectors that can be then fed to machine learning models in a classical way

Types

Machine learning algorithms cannot work with raw text directly; the text must be converted into numbers. Specifically, vectors of numbers.



N-grams

N-grams are the combination of multiple words used together. Ngrams with N=1 are called unigrams. Similarly, bigrams (N=2), trigrams (N=3) and so on can also be used.

This is Big Data AI Book

Bag of Words (BoW)

- used to analyze text and documents based on **word count.**
- model does not account for word order within a document.

	about	bird	heard	is	the	word	you
About the bird, the bird, bird bird bird	1	5	0	0	2	0	0
You heard about the bird	1	1	1	0	1	0	1
The bird is the word	0	1	0	1	2	1	0

Bag of Words(BOW) Limitation

'The sky is blue and beautiful',
'The king is old and the queen is beautiful',
'Love this beautiful blue sky',
'The beautiful queen and the old king']

	and	beautiful	blue	is	king	love	old	queen	sky	the	this
0	1	1	1	1	0	0	0	0	1	1	0
1	1	1	0	2	1	0	1	1	0	2	0
2	0	1	1	0	0	1	0	0	1	0	1
3	1	1	0	0	1	0	1	1	0	2	0

	beautiful	beautiful blue	beautiful queen	blue	blue beautiful	blue sky	king	king old	love	love beautiful	old	old king	old queen	queen	queen beautiful	queen old	sky	sky blue
0	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	1
1	1	0	0	0	0	0	1	1	0	0	1	0	1	1	1	0	0	0
2	1	1	0	1	0	1	0	0	1	1	0	0	0	0	0	0	1	0
3	1	0	1	0	0	0	1	0	0	0	1	1	0	1	0	1	0	0

Term Document – Inverse Document Frequnecy Matrix

The *term frequency* is a ratio of the count of a word's occurrence in a document and the number of words in the document

Let us show the count of word i in document j by tf_{ij}

Let us represent document frequency for word i by df_i. With N as the number of documents in the corpus, the tf-idf weight w_{ij} for word i in document j is computed by the following formula:

The *document frequency* of word i represents the number of documents in the corpus with word i in them

$$w_{i,j} = tf_{i,j} \times log(\frac{N}{df_i})$$

TF-IDF Calculation Example

Words	Count		Term Frequency (TF)		Inverse Document Frequency (IDF)	TF * IDF	
	Document 1	Document 2	Document 1	Document 2		Document 1	Document 2
read	1	1	0.17	0.17	0	0	0
svm	1	0	0.17	0	0.3	**0.05**	0
algorithm	1	1	0.17	0.17	0	0	0
article	1	1	0.17	0.17	0	0	0
dataaspirant	1	1	0.17	0.17	0	0	0
blog	1	1	0.17	0.17	0	0	0
randomforest	0	1	0	0.17	0.3	0	**0.05**

Text representation

The quick brown fox jumped over the brown dog

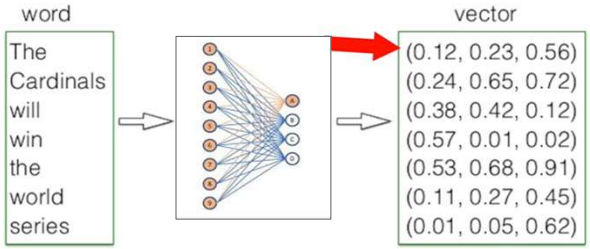
One-hot Embedding

cat	the	quick	brown	fox	jumped	over	dog	bird	flew	...	kangaroo	house
0	1	0	0	0	0	0	0	0	0	...	0	0
0	0	1	0	0	0	0	0	0	0	...	0	0
0	0	0	1	0	0	0	0	0	0	...	0	0
0	0	0	0	1	0	0	0	0	0	...	0	0
0	0	0	0	0	1	0	0	0	0	...	0	0
0	0	0	0	0	0	1	0	0	0	...	0	0
0	1	0	0	0	0	0	0	0	0	...	0	0
0	0	0	1	0	0	0	0	0	0	...	0	0
0	0	0	0	0	0	0	1	0	0	...	0	0

Dictionary Size

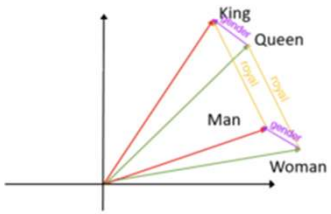
Word Embeddings

$$\underset{\mathbb{R}^{m\times n}}{W}\ \underset{\mathbb{R}^{n\times 1}}{x} + \underset{\mathbb{R}^{m\times 1}}{b}$$



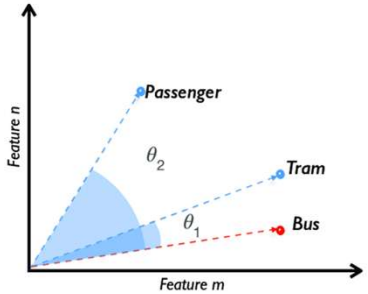
Word Embeddings

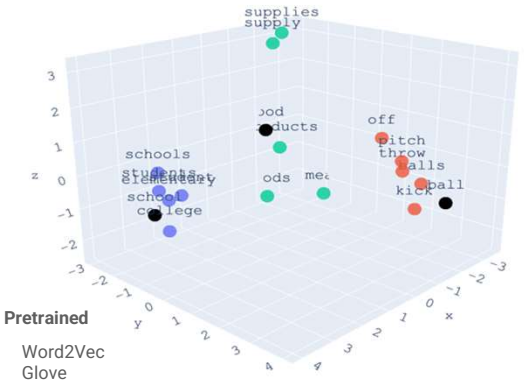
Semantic space captures word relationships



King - Man + Woman = ???

Cosine distance reflects semantic similarity





How do we achieve this ?

"Distributional Semantics"

Pretrained

Word2Vec
Glove
FastText

Distributional Word Representation

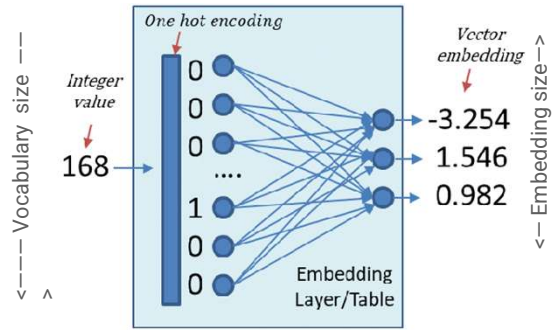
You shall know a word by the company it keeps (Firth, J. R. 1957:11).

A bottle of tezgüino is on the table.
Everyone likes tezgüino.
Tezgüino makes you drunk.
We make tezgüino out of corn.

Can you understand what tezgüino means ?

Image courtesy: **Lena Voita's NLP course | For You** https://lena-voita.github.io/nlp_course.html

Neural Word Embeddings



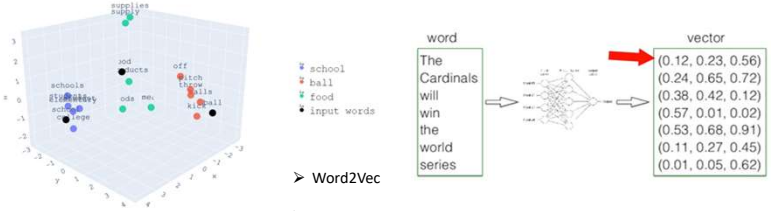
Word Embedding

This is a word embedding for the word "king" (GloVe vector trained on Wikipedia):

```
[ 0.50451 , 0.68607 , -0.59517 , -0.022801, 0.60046 , -0.13498 , -0.08813 , 0.47377 , -
0.61798 , -0.31012 , -0.076666, 1.493 , -0.034189, -0.98173 , 0.68229 , 0.81722 , -0.51874
, -0.31503 , -0.55809 , 0.66421 , 0.1961 , -0.13495 , -0.11476 , -0.30344 , 0.41177 , -
2.223 , -1.0756 , -1.0783 , -0.34354 , 0.33505 , 1.9927 , -0.04234 , -0.64319 , 0.71125 ,
0.49159 , 0.16754 , 0.34344 , -0.25663 , -0.8523 , 0.1661 , 0.40102 , 1.1685 , -1.0137 , -
0.21585 , -0.15155 , 0.78321 , -0.91241 , -1.6106 , -0.64426 , -0.51042 ]
```

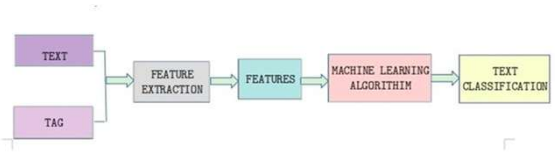
Text Embedding

Word Embedding is the representation of text in the form of vectors. The underlying idea here is that similar words will have a minimum distance between their vectors.



- Word2Vec
- Doc2Vec

Text Classification -Pipeline



Thank You

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ANY QUESTION